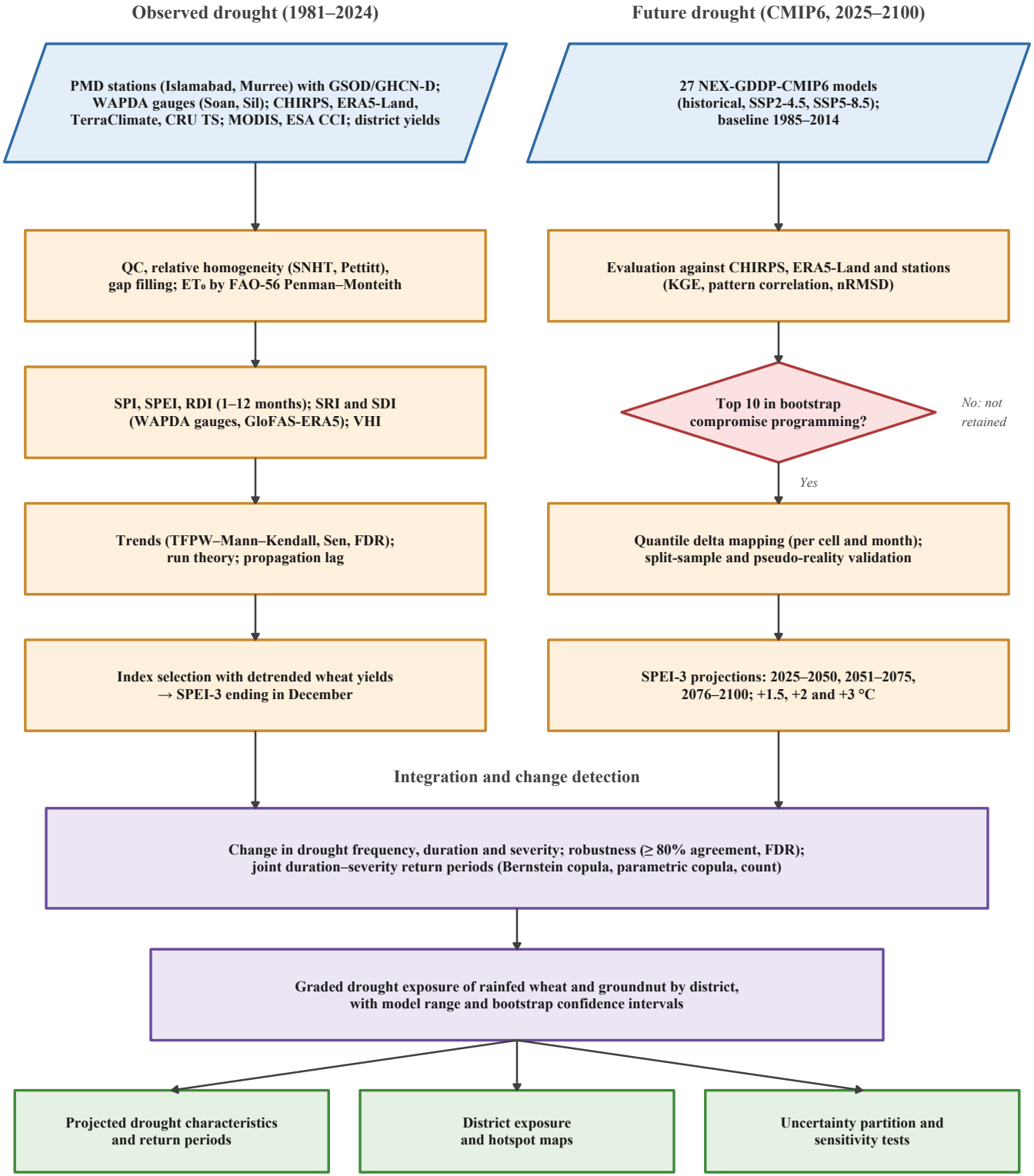
Observed drought (1981–2024)
Future drought (CMIP6, 2025–2100)
PMD stations (Islamabad, Murree) with GSOD/GHCN-D; WAPDA gauges (Soan, Sil); CHIRPS, ERA5-Land, TerraClimate, CRU TS; MODIS, ESA CCI; district yields
27 NEX-GDDP-CMIP6 models (historical, SSP2-4.5, SSP5-8.5); baseline 1985–2014
QC, relative homogeneity (SNHT, Pettitt), gap filling; ET₀ by FAO-56 Penman–Monteith
Evaluation against CHIRPS, ERA5-Land and stations (KGE, pattern correlation, nRMSD)
SPI, SPEI, RDI (1–12 months); SRI and SDI (WAPDA gauges, GloFAS-ERA5); VHI
Top 10 in bootstrap compromise programming?
No: not retained
Yes
Trends (TFPW–Mann–Kendall, Sen, FDR); run theory; propagation lag
Quantile delta mapping (per cell and month); split-sample and pseudo-reality validation
Index selection with detrended wheat yields → SPEI-3 ending in December
SPEI-3 projections: 2025–2050, 2051–2075, 2076–2100; +1.5, +2 and +3 °C
Integration and change detection
Change in drought frequency, duration and severity; robustness (≥ 80% agreement, FDR); joint duration–severity return periods (Bernstein copula, parametric copula, count)
Graded drought exposure of rainfed wheat and groundnut by district, with model range and bootstrap confidence intervals
Projected drought characteristics and return periods
District exposure and hotspot maps
Uncertainty partition and sensitivity tests